Greater the frequency of the incident radiation, larger is the corresponding stopping potential. This implies that *as the frequency is increased, the photoelectrons are emitted with greater kinetic energies so that the retarding potential needed to stop the photoelectrons is also greater.*

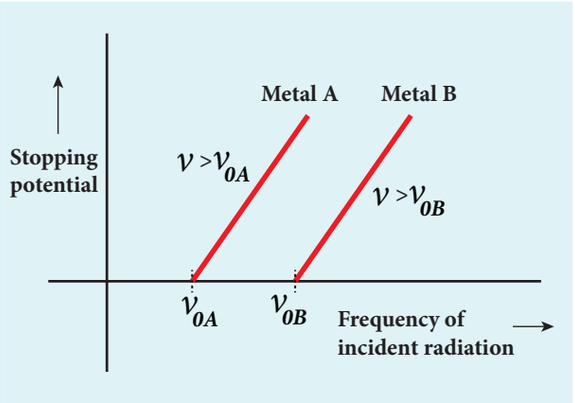


Figure 8.12 Variation of stopping potential with frequency of the incident radiation for two metals

Now a graph is drawn between frequency of incident radiation and the stopping potential for different metals (Figure 8.12). From this graph, it is found that stopping potential varies linearly with frequency. Below a certain frequency called threshold frequency, no electrons are emitted; hence stopping potential is zero for that reason. But as the frequency is increased above threshold value, the stopping potential varies linearly with the frequency of incident light.

8.2.5 Laws of photoelectric effect

The above detailed experimental investigations of photoelectric effect revealed the following results:

i) For a given metallic surface, the emission of photoelectrons takes place only if the frequency of incident light is greater than a certain minimum frequency called the threshold frequency.

ii) For a given frequency of incident light (above threshold value), the number of photoelectrons emitted is directly proportional to the intensity of the incident light. The saturation current is also directly proportional to the intensity of incident light.

iii) Maximum kinetic energy of the photo electrons is independent of intensity of the incident light.

iv) Maximum kinetic energy of the photo electrons from a given metal is directly proportional to the frequency of incident light.

v) There is no time lag between incidence of light and ejection of photoelectrons.

Once photoelectric phenomenon has been thoroughly examined through various experiments, the attempts were made to explain it on the basis of wave theory of light.

8.2.6 Concept of quantization of energy

Failures of classical wave theory

From Maxwell's theory (Refer unit 5 of volume 1), we learnt that light is an electromagnetic wave consisting of coupled electric and magnetic oscillations that move with the speed of light and exhibit typical wave behaviour. Let us try to explain the experimental observations of photoelectric effect using wave picture of light.

i) When light is incident on a metallic surface, there is a continuous supply of energy to the electrons in the metal surface. According to wave theory, light of greater intensity should impart greater kinetic energy to the liberated electrons (Here, Intensity of light is the energy delivered per unit area per unit time).

XII_Physics_Vol-2_UNIT-8_EM.indd 116 03-03-2022 15:20:24